MSMS 408 : Practical 17

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Question

(i) Generate a sample of size 1000 from a mixture of Gaussians given by the following PDF.

$$f(x) = 0.6 \cdot \mathcal{N}(0,1) + 0.4 \cdot \mathcal{N}(4,1).$$

Each x_i is generated from $\mathcal{N}(0,1)$ with probability 0.6 or from $\mathcal{N}(4,1)$ with probability 0.4. Assume that you are given only x_i's and you don't know which of the two Gaussians each x_i comes from.

(ii) Use Expectation-Maximization algorithm to estimate the mixing proportions, ϕ_1, ϕ_2, say; the Gaussian means, μ_1, μ_2, say and the Gaussian variances σ_1^2, σ_2^2, say.

Theory

1 Data simulation :

- Generate $u_1, \ldots, u_n \sim U(0,1)$.
- Set

$$\forall i = 1(1)n,\ x_i = \begin{cases} x \sim \mathcal{N}(0,1) & \text{if } u_i <= 0.6 \\ x \sim \mathcal{N}(4,1) & \text{if } u_i > 0.6 \end{cases}.$$

2 EM Algorithm :

- $\boldsymbol{\theta} = \left\{\phi_1, \phi_2, \mu_1, \mu_2, \sigma_1^2, \sigma_2^2\right\}$.
- Start with an initial value $\boldsymbol{\theta}^{(0)}$.
- Repeat until convergence {

(E-Step) For each i, j :

$$w_j^{(i)} := p\left(z_j^{(i)} = 1 \mid x^{(i)}, \boldsymbol{\theta}^{(t)}\right)$$

(M-Step) Update :

$$\phi_j := \frac{1}{n} \sum_{i=1}^{n} w_j^{(i)},$$

$$\mu_j := \frac{\sum_{i=1}^{n} w_j^{(i)} x^{(i)}}{\sum_{i=1}^{n} w_j^{(i)}},$$

$$\sigma_j^2 := \frac{\sum_{i=1}^{n} w_j^{(i)} \left(x^{(i)} - \mu_j\right)^2}{\sum_{i=1}^{n} w_j^{(i)}}$$

}.

R Program

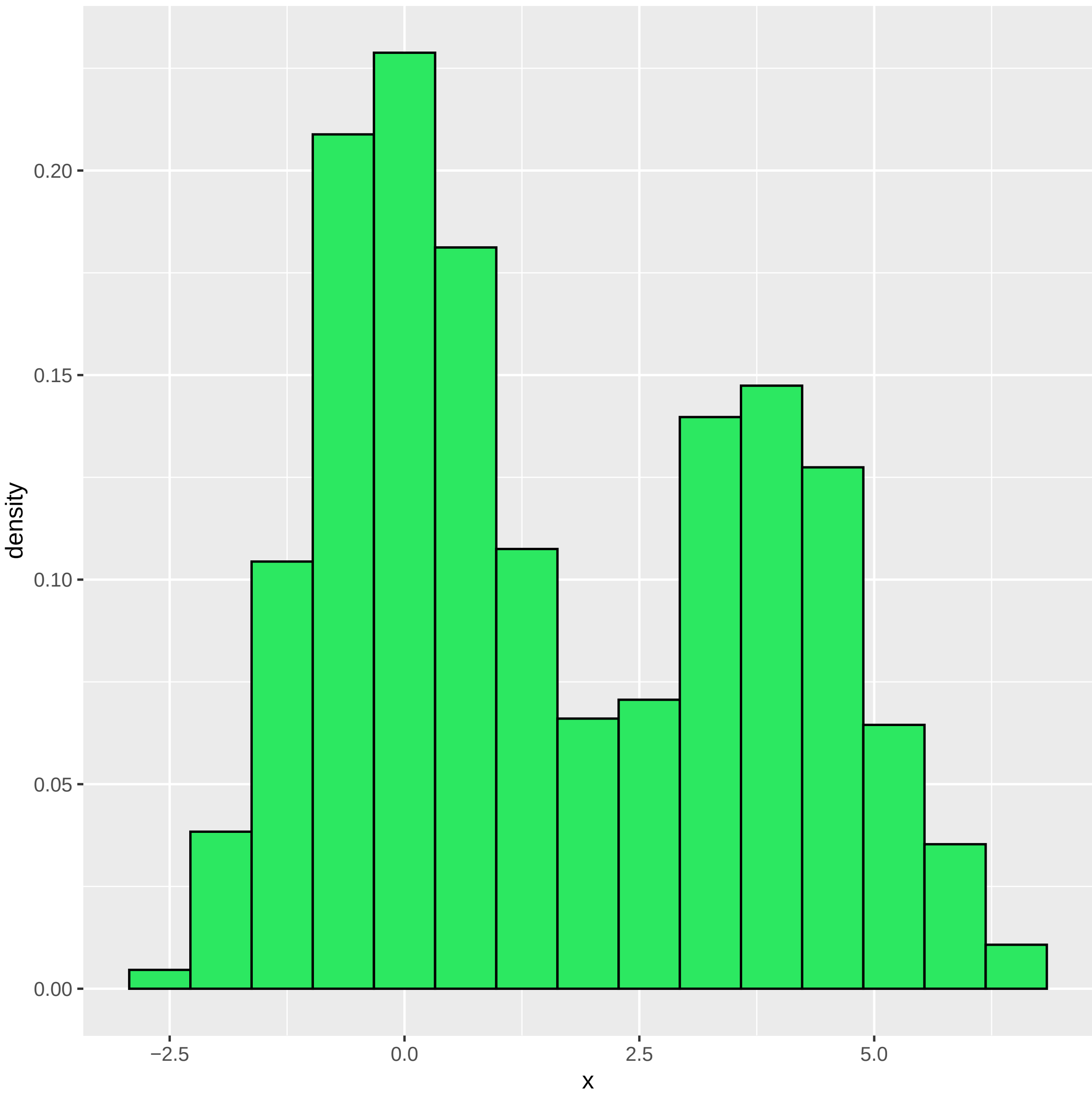
1

```r
n <- 1000

u <- runif(n)

theta.true <- c(0.6, 0.4, 0, 4, 1, 1)

x <- ifelse(u < 0.6, rnorm(n, mean = 0, sd = 1), rnorm(n, mean = 4, sd = 1))
```

Here is a histogram of the generated sample.

```r
library(tidyverse)
```

```r
data.frame(x) %>%
  ggplot(aes(x = x)) +
  geom_histogram(aes(y = after_stat(density)), bins = 15,
                 color = "black", fill = "#2CE861")
```



✎ Clearly the sample is from a bimodal distribution, each peak governed by a Gaussian.

2 From the histogram, 0 and 3 seem reasonable initialization of locations.

```
theta <- c(0.4, 0.6, 0, 3, 2, 2)
```

```
epsilon <- 0.0001
```

```
repeat{
  # current setting
  phi1 <- theta[1]; phi2 <- theta[2]
  mu1 <- theta[3]; sigma1.sq <- theta[5]
  mu2 <- theta[4]; sigma2.sq <- theta[6]

  # E-Step
  joint.prob <- matrix(NA, nrow = n, ncol = 2)
  joint.prob[,1] <- dnorm(x, mu1, sqrt(sigma1.sq)) * phi1
  joint.prob[,2] <- dnorm(x, mu2, sqrt(sigma2.sq)) * phi2

  marginal.prob <- rowSums(joint.prob)

  posterior.prob <- joint.prob / marginal.prob # using recycling property of R

  # M-Step

  theta.new <- c()

  # update phi
  theta.new[c(1, 2)] <- colMeans(posterior.prob)

  # update mu
  theta.new[c(3, 4)] <- colSums(posterior.prob * x) / colSums(posterior.prob)

  # update sigma.sq
  temp <- (x - matrix(rep(c(mu1, mu2), n), nrow = n, ncol = 2, byrow = TRUE))^2

  theta.new[c(5, 6)] <- colSums(posterior.prob * temp) / colSums(posterior.prob)

  if(max(abs(theta - theta.new)) < epsilon){
    theta <- theta.new
    break
  }else{
    theta <- theta.new
  }
}
```

```
theta

## [1]  0.5808551  0.4191449 -0.0603533  3.8633547  0.8836203  1.1900673
```

Conclusion

Parameter	Estimate	True Value
$\widehat{\phi_1}$	0.581	0.6
$\widehat{\phi_2}$	0.419	0.4
$\widehat{\mu_1}$	-0.06	0
$\widehat{\mu_2}$	3.863	4
$\widehat{\sigma_1^2}$	0.884	1
$\widehat{\sigma_2^2}$	1.19	1

Table 1: Estimated and true parameter values

✎ Even though EM algorithm has a tendency to converge to the local optima, it has performed considerably well in our problem.